End-to-End Testing — HelixVPN nano-detail spec (Volume 8 · §11.4.169 type 3)

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> **Reconciled (§11.4.35, 2026-06-26):** the §2 harness topology + skeletons were rewritten onto the **canonical four-namespace rig** (`client` / `censor` / `gateway` / `exit`) of test-rig.md, with the DPI/censor in a distinct `censor-ns` middlebox and the wire/DPI capture point on the censor egress `cen2gw`. The earlier inferior 3-namespace rig (`client` / `gateway` / `lanA` with the `nft` block bolted onto the gateway forward hook) is superseded — test-rig.md is the single source for the rig's namespace/interface names and capture points.

> Nano-detail expansion of §5.3 of the Volume-8 overview. E2E proves the **whole reachability slice on one box**: a real control plane issues a NetworkMap, a real `helix-core` client builds a real tunnel, real packets cross `client → gateway → connector LAN`, and a real `curl` returns the hello page — or, for the negative E2E, a SYN leaves and **zero** SYN-ACK returns (default-deny). The substrate is the Linux **netns + nftables-DPI + tc-netem rig** [04_P0 §3]. This is where bluff-classes **B1 (config-only)** and **B3 (wrong-plane)** are mechanically defeated by pcap + body-hash + `wg`-counter-advance evidence (§11.4.107). Spec-only; unproven assumptions marked `UNVERIFIED`. Siblings: unit.md, integration.md, full-automation.md, challenges.md, helixqa.md.

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1. Scope — the reachability slices E2E covers

E2E owns the user-visible security properties that **only a packet on a real tunnel can prove**. Each slice maps to a Phase-1 Definition-of-Done AC (overview §7.2).

Slice	Path	The user-visible property	Bound AC
Authorized reachability	client → gateway → connector LAN, `curl http://10.10.0.20/` ⇒ 200	the user reaches a host the policy grants	**AC2**
Default-deny (negative E2E)	client → gateway, SYN to an unauthorized host	a host the policy does **not** grant is unreachable — SYN out, **zero** SYN-ACK	**AC3**
Transport escalation	plain-WG UDP blocked → auto-escalate to MASQUE/ H3 → reach preserved	the tunnel survives a DPI UDP block, classified as HTTP/3 not WireGuard	**AC4**
Policy reconcile (live)	a policy edit reconfigures the running slice with **no restart**	a revoked/added grant takes effect on the live tunnel	**AC5**
Revoke (live)	revoke a device → its peer drops from every map → tunnel sealed	a revoked device loses reachability	**AC6**
Resilience		the tunnel holds goodput under	G1/G2 [04_P0]

Slice	Path	The user-visible property	Bound AC
	reach under 5% loss / 40 ms jitter (tc-netem)	impairment (adjacency to STRESS §11.4.85)	

The rig is the **reproducible substrate** for AC2/AC3/AC4 and survives Phase-0→1 (overview §5.3). E2E does **not** drive the Flutter UI — that is UI/REC via `panoptic` (overview §5.14); E2E drives the *core* and the *wire*.

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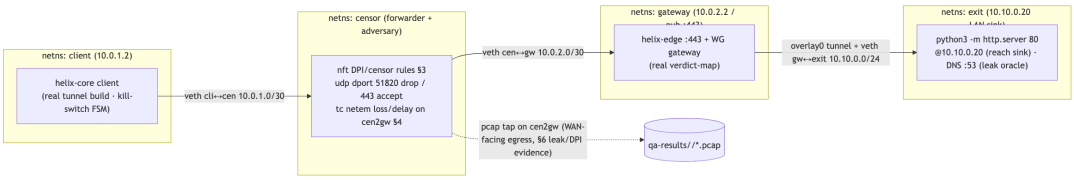
2. Harness — the netns + nftables-DPI + tc-netem rig

The E2E substrate is the **canonical four-namespace rig** defined in test-rig.md: four real Linux network namespaces — `client`, `censor`, `gateway`, `exit` — on one host. The adversary is split out of the path as its **own `censor-ns` middlebox** the packet must traverse, so the DPI/censor `nft` ruleset is a *distinct forwarding hop* (not an `nft` rule bolted onto an endpoint, test-rig.md §0); `tc netem` on the censor egress (`cen2gw`) injects loss/delay for the resilience slice. The rig is the Phase-0 deliverable HVPN-P0-022/023 (06). The rig scripts (`rig/netns_up.sh`, `rig/netns_down.sh`, the regime `nft` profiles, `rig/impair.sh`, `rig/exercise.sh`) are defined canonically in test-rig.md §§2–4 and shared by every skeleton below — same namespace names, same interface names, same `cen2gw` capture point.



The `sudo` for `ip netns add` is the **only** privileged step in the whole harness and is a documented, scoped `CAP_NET_ADMIN` exception — never a container-management escalation (overview §9, §11.4.161). Container infra (the real control plane's PG/Redis behind the gateway) is still booted rootless via `containers` (integration.md §2).

```
# rig/netns_up.sh — the 4-ns rig (CANONICAL definition: test-
        rig.md §2)
#   client(10.0.1.2) -> censor -> gateway(10.0.2.2:443) ->
        exit(10.10.0.20)
set -euo pipefail
for ns in client censor gateway exit; do ip netns add "$ns" 2>/
        dev/null || true; done
# veths client↔censor (10.0.1.0/30) · censor↔gateway (10.0.2.0/30)
        · gateway↔exit (10.10.0.0/24),
```

```
# forwarding + routes, and the exit-ns sinks are wired in full by
        test-rig.md §2:
ip netns exec exit python3 -m http.server 80 --bind 10.10.0.20
        &   # the hello service (sink @10.10.0.20)

# rig/dpi_block.sh — DPI/censor sim on the CENSOR-ns forward hook
        (the middlebox), AC4 escalation E2E
ip netns exec censor nft add table inet censor
ip netns exec censor nft add chain inet censor fwd '{ type filter
        hook forward priority 0; policy accept; }'
ip netns exec censor nft add rule  inet censor fwd udp dport 51820
        drop   # kill plain WireGuard
ip netns exec censor nft add rule  inet censor fwd udp dport 443
        accept  # allow MASQUE/H3

# rig/impair.sh — loss/jitter on the censor egress cen2gw
        (§11.4.85 adjacency; CANONICAL: test-rig.md §4)
ip netns exec censor tc qdisc add dev cen2gw root netem loss 5%
        delay 40ms 10ms
```

Cleanup is non-negotiable: `trap 'rig/netns_down.sh' EXIT` deletes all four namespaces and leaves the host quiescent (§11.4.14); the orchestrator's post-test sanity check FAILs the run if any `client`/`censor`/`gateway`/`exit` namespace, the `inet censor` `nft` table, or a `netem` qdisc survives.

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3. Fixtures — real processes, simulated network, zero mocks

Fixture	Real or mock	Notes
`helix-core` client	**real build**	builds a real WireGuard tunnel (boringtun or kernel WG, ../v02-data-plane/wireguard-core.md)
gateway + `helix-edge`	**real build**	real verdict-map packet classification
control plane (PG/Redis/coordinator)	**real**, booted via `containers`	issues the real NetworkMap the client applies
connector LAN sink	**real** `python3 -m http.server`	a real HTTP server — the reachability target
DPI block	**real** `nftables` rule	a real kernel drop of UDP/51820 (not a mock)
network impairment	**real** `tc netem`	real loss/jitter in the kernel
mocks	**FORBIDDEN** (§11.4.27)	E2E is post-unit; a mock socket here is a §11.4.27 violation

Everything on the path is real; only the *network topology* is simulated (netns + nftables + netem stand in for "a censored hostile network" reproducibly). This is exactly the §11.4.3 per-environment-topology dispatch: the rig **is** the topology, and absence of `CAP_NET_ADMIN` ⇒ honest `SKIP: topology_unsupported`, never a fake PASS.

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4. Evidence taxonomy + the §11.4.107 liveness battery

An E2E PASS cites **packet-level captured evidence** (§11.4.69 classes `network_throughput` / `network_connectivity`), never a status field (`B3`) or a config check (`B1`).

Slice	§11.4.69 evidence shape	Defeats
authorized reach (AC2)	pcap with a SYN→**SYN-ACK** from 10.10.0.20 **and** a non-empty `curl` body hash	B1 (config-only)
default-deny (AC3)	pcap showing SYN out, **zero** SYN-ACK; edge verdict-map `drop++` counter	B1, B2
escalation (AC4)	`StatusReport.transport=="masque-h3"` **and** tshark classifies the flow as HTTP/3 with **no** WG signature	B3 (wrong-plane)
reach liveness (G1)	`iperf3 -J` JSON `sum_received.bits_per_second > 0` over ⩾10 s **and** `wg show <if> transfer` rx/tx advancing	B3
resilience (G2)	goodput under `netem` ⩾ budget; pcap of decrypted inner packets	B3

The §11.4.107 liveness battery, instantiated for E2E (overview §3.2):

1. **Liveness over a window, not a point** — throughput sampled ⩾10 s; a one-shot `ping` reply is a pre-filter, never the proof (§11.4.107(1)).
2. **Independent counter-advance** — goodput (`iperf3`) **and** the kernel WG counter (`wg show ... transfer`) must *both* move; a flat WG counter with moving `iperf3` means a decoy/loopback path → FAIL (§11.4.107(2)).
3. **Not-stale cross-check** — after a transport escalation, the post-escalation pcap's first inner packet must not replay the pre-escalation flow's last packet (§11.4.107(4)).
4. **Loading is a distinct state** — a handshake in progress is `Connecting`, not `Connected`; liveness is judged only after the FSM reaches `Connected` (§11.4.107(3)); timeout+reachable ⇒ FAIL, timeout+unreachable ⇒ SKIP.

5. **Metamorphic relations** (no golden source) — the same LAN host reached over WG-UDP vs MASQUE returns the **same body hash**; a paused tunnel stops the WG counter; 2× clients ⇒ ~2× aggregate goodput (§11.4.107(8)).

Every analyzer (the tshark classifier, the leak detector) is **self-validated** (§11.4.107(10)) with a golden-good + golden-bad pcap pair (overview §3.3) — defeating bluff-class **B5**.

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5. Determinism — N-iteration identical evidence

Per §11.4.50, E2E runs N=3 (normal) / N=10 (cycle-validation) and must produce identical *verdicts* with structurally-identical evidence:

1. **Body-hash equality** — the `curl` body hash is identical across runs (the sink serves a fixed page); a diverging hash is auto-FAIL.
2. **Verdict equality** — pass→pass→pass or the run is a flake-FAIL; "first reach worked, retry it" is forbidden (§11.4.50 mechanises §11.4.7).
3. **Deterministic impairment seed** — `tc netem` uses a fixed loss/delay so the resilience slice's goodput stays within a tolerance band across runs; the band, not an exact value, is the assertion.
4. **Clean rig per iteration** — `netns_up.sh`/`netns_down.sh` bracket each iteration so no stale tunnel/route/qdisc shadows the next run (§11.4.108/.139 clean-artifact; defeats B4 stale-state).

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6. Acceptance gate — when E2E blocks a release

Gate	Bar	Bound item
`make e2e` (overview §9)	positive reach + negative deny both GREEN with pcap	AC2 + AC3
G1 (Phase-0)	`iperf3` ⩾ **80%** bare-link; `curl 10.10.0.20:80` OK	E2E + BENCH
G2 (Phase-0)	core over MASQUE through a DPI UDP block ⩾ **50%** of plain; tshark = HTTP/3, no WG sig; survives 5% loss	E2E + SEC
G6 (Phase-0)	a static-map edit reconciles the slice with **no restart**	E2E + INT
AC4 / AC5 / AC6 (Phase-1)	escalation / reconcile / revoke proven on the live tunnel	E2E challenges

E2E is the **highest-authority reachability gate**; an E2E FAIL is a release blocker and triggers the §11.4.4 STOP + §11.4.114 known-good isolation. Per §11.4.132 risk-ordering the **security-floor E2E** (default-deny AC3, revoke AC6) runs *before* any convenience slice, and only after that set is GREEN with captured pcap does the rest run.

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7. The paired §1.1 mutation (anti-bluff proof)

```
# §1.1 mutation for CM-DEFAULT-DENY-E2E (defeats the most
dangerous bluff: fail-open)
- in helix-edge verdict-map, change the default action:
    default: DROP                 // correct: deny unless granted
    MUTATED: default: ACCEPT      // fail-OPEN — every host
reachable
- assert: rig/test_reach.sh deny now sees a SYN-ACK from the
unauthorized host → E2E FAILs
- restore; assert deny PASS (zero SYN-ACK) again (§11.4.84
quiescence)

# §1.1 mutation for CM-ESCALATION-E2E (defeats B3 wrong-plane)
- force the escalation path to leave the StatusReport.transport at
"wg-udp" while the
  DPI block is active (the bug: reports masque but actually still
tries WG)
- assert: tshark finds NO HTTP/3 flow + reach FAILs → the test
catches the lie
```

The leak/classifier analyzers' golden-bad fixtures (a seeded-leak pcap, a WG flow mislabelled as H3) **must** FAIL their analyzers — the §11.4.107(10) self-validation that defeats B5; stripping the discriminating tshark filter makes golden-bad pass → the meta-test catches the mutation.

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8. Test skeletons

8.1 The reach assertion — positive + negative (the core E2E)

```
# rig/test_reach.sh — §11.4.50 N=3 deterministic; sources the
        anti-bluff lib (§11.4.1)
run_reach() {                                             # $1 =
        expected: pass|deny
  pcap="qa-results/e2e/$(date +%s)_reach.pcap"
  ip netns exec client tcpdump -i overlay0 -w "$pcap" &  TPID=$!
  body=$(ip netns exec client curl -s --max-time 5 http://
        10.10.0.20/ || true)
  kill "$TPID" 2>/dev/null
  if [ "$1" = pass ]; then
```

```
    [ -n "$body" ] || ab_fail "B1: no body — config-only PASS
        forbidden"
    tshark -r "$pcap" -Y 'tcp.flags.syn==1 && tcp.flags.ack==1' |
        grep -q . \
      && ab_pass_with_evidence "authorized reach" "$pcap" \
      || ab_fail "no SYN-ACK captured — reach not proven on the
        wire"
  else
    [ -z "$body" ] || ab_fail "default-deny breached: got a body"
    tshark -r "$pcap" -Y 'tcp.flags.syn==1 && tcp.flags.ack==1' |
        grep -q . \
      && ab_fail "deny breached: SYN-ACK present" \
      || ab_pass_with_evidence "default-deny (no SYN-ACK)" "$pcap"
  fi
}
```

8.2 Liveness with independent counter-advance (defeats B3)

```
# rig/liveness.sh — both iperf3 goodput AND the kernel WG counter
        must advance (§11.4.107(2))
wg_rx_before=$(ip netns exec client wg show overlay0 transfer |
        awk '{print $2}')
THRPUT=$(ip netns exec client iperf3 -J -t 10 -c 10.10.0.20 | jq
        '.end.sum_received.bits_per_second')
wg_rx_after=$(ip netns exec client wg show overlay0 transfer |
        awk '{print $2}')
adv=$(( wg_rx_after - wg_rx_before ))
{ awk -v t="$THRPUT" 'BEGIN{exit !(t>0)}' && [ "$adv" -gt 0 ]; } \
  && ab_pass_with_evidence "live tunnel: goodput $THRPUT, wg+$adv
        bytes" "qa-results/e2e/liveness_$(date +%s)/" \
  || ab_fail "decoy path: iperf=$THRPUT but wg counter advanced
        $adv (B3 wrong-plane)"
```

8.3 Transport escalation through a DPI block (AC4)

```
# rig/escalate.sh — plain WG dropped → core auto-escalates to
        MASQUE/H3, reach preserved
rig/dpi_block.sh
        # nft drops udp/51820, accepts udp/443
ip netns exec client helix-core connect --target 10.10.0.20 --
        status-json /tmp/status.json
sleep 6
        # allow the ladder to escalate (FSM → Connected)
transport=$(jq -r '.transport' /tmp/status.json)
[ "$transport" = "masque-h3" ] || ab_fail "did not escalate:
        transport=$transport (B3)"
pcap="qa-results/e2e/$(date +%s)_escalate.pcap"
ip netns exec censor tcpdump -i cen2gw -w "$pcap" & TP=$!   # tap
        the censor↔gateway path where DPI classifies (test-rig.md
        §7)
ip netns exec client curl -s --max-time 5 http://10.10.0.20/ >/
        dev/null
kill "$TP" 2>/dev/null
tshark -r "$pcap" -Y 'http3' | grep -q . \
```

```
  && ! tshark -r "$pcap" -Y 'wg' | grep -q . \
  && ab_pass_with_evidence "AC4: escalated to MASQUE/H3, no WG signature" "$pcap" \
  || ab_fail "escalation evidence missing (H3 absent or WG signature present)"
```

8.4 Live revoke (AC6) — peer drops from the running tunnel

```
# rig/revoke.sh — revoke a device, assert the live tunnel is sealed < 1 s (SLO3 budget)
t0=$(date +%s.%N)
helixvpnctl revoke --device device-a1                  # control-plane revoke
# poll the live edge until the peer is removed (loading is distinct, §11.4.107(3))
until ! ip netns exec gateway wg show overlay0 peers | grep -q "$DEV_A1_PUBKEY"; do
  sleep 0.05; over_budget 1.0 "$t0" && ab_fail "revoke > 1s (SLO3 missed)"
done
t1=$(date +%s.%N); echo "$t0,$t1" >> qa-results/e2e/revoke_timing.csv
# and the revoked device can no longer reach the LAN host:
ip netns exec client curl -s --max-time 3 http://10.10.0.20/ | grep -q . \
  && ab_fail "revoked device still reaches the LAN host" \
  || ab_pass_with_evidence "AC6: revoke enforced < 1s" "qa-results/e2e/revoke_timing.csv"
```

8a. The rig's four-layer mapping (§11.4.108) + clean-artifact discipline

Every E2E slice crosses the §11.4.108 four layers; the rig makes each layer's assertion explicit so a SOURCE-green-but-RUNTIME-dead bug cannot pass (the §11.4.108 forensic class):

Layer	E2E assertion	Defeats
SOURCE	the policy/verdict-map source compiles to the expected routes (UNIT, unit.md §8.1)	—
ARTIFACT	the deployed edge binary's verdict-map **bytes** carry the route (not just the source)	B4 stale-state
RUNTIME-on-clean-target	a **fresh** `netns_up.sh` rig, no stale tunnel/route, runs the slice	B4
USER-VISIBLE		B1, B3

Layer	E2E assertion	Defeats
	the pcap shows the SYN-ACK / the absence thereof	

Clean-artifact discipline (§11.4.108/.139, defeats B4): the rig is brought down (`netns_down.sh`) and up fresh before each E2E iteration so validation never runs against a previous deploy's running daemon (a stale tunnel, a leftover route, an old verdict-map). The runtime-signature for an E2E slice is "the captured pcap on a freshly-deployed clean rig shows the expected wire behaviour" — a pcap captured against a stale rig is **invalid** and any PASS it produces is a §11.4 PASS-bluff.

8b. Bluff-class guard map (the five forbidden E2E bluffs)

Bluff	What it would look like in E2E	The rig assertion that forbids it
B1 config-only	"AllowedIPs contains the route" with no packet	`[ -n "$body" ]` + SYN-ACK pcap row required (§8.1)
B2 absence-of-error	"kill-switch enabled, no error"	`LEAK==0 && DNS==0` packet counts on a real drop (SEC §5.7)
B3 wrong-plane	`transport=="masque-h3"` but WG on the wire	independent `wg`-counter-advance + tshark H3-present/WG-absent (§8.2/§8.3)
B4 stale-state	validating an old running daemon	fresh `netns_up.sh` per iteration (§8a)
B5 unvalidated analyzer	the tshark classifier passes its golden-bad	self-validated golden-good/bad pcap pair (§11.4.107(10), §4)

9. Open decisions surfaced for QA

#	Decision	Options	Recommendation
E-D1	WG datapath in the rig	kernel WG vs boringtun userspace	**match the platform-under-test** (../v02-data-plane/wireguard-core.md); run both where both ship, evidence per-datapath (§11.4.3)

#	Decision	Options	Recommendation
E-D2	DPI sim fidelity	nftables UDP-port drop (simple) vs a real DPI engine	**nftables port-drop for MVP** (proves AC4 escalation); a real DPI engine is a Phase-2 SEC deepening (`UNVERIFIED` whether GFW-class active probing is reproducible locally)
E-D3	rig host	local dev box vs the parked `nezha.local` heavy node (../v06-deploy/remote-testing-infra.md)	**local for MVP**; remote node PARKED until greenlit — the rig is host-portable by design

> **`UNVERIFIED`:** the exact `StatusReport.transport` enum value `"masque-h3"` and the `wg show ... transfer` output column index are taken from the overview skeleton; confirm against the FFI surface (../v04-client/ffi-surface.md) and the deployed `wg(8)` version at rig build time.

Sources verified

- [Volume-8 overview](#) §2 (taxonomy row `E2E`), §3.1/§3.2/§3.3 (evidence taxonomy + liveness battery + self-validated analyzers), §5.3 (the rig scripts), §7.1 (G1/G2/G6), §7.2 (AC2–AC6), §9 (Makefile `e2e:`) — read 2026-06-26.
- Sibling specs cross-referenced: ../v02-data-plane/wireguard-core.md, ../v02-data-plane/transport-masque-quic.md, ../v02-data-plane/transport-selection-ladder.md, ../v02-data-plane/routing-and-addressing.md, ../v03-control-plane/reconciliation-flow.md, ../v04-client/ffi-surface.md, ../v06-deploy/remote-testing-infra.md — filenames confirmed present 2026-06-26 (v06 doc is `[GEN]`-planned per MASTER_INDEX).
- Constitution: §11.4.107 (liveness / no-frozen-frame / self-validated analyzer / metamorphic), §11.4.3 (topology dispatch / SKIP), §11.4.5/.68/.69 (captured / no-absence-of-error / sink-side evidence), §11.4.50 (determinism), §11.4.108/.139 (clean-artifact runtime signature), §11.4.114 (known-good isolation), §11.4.132 (risk-order), §11.4.14 (cleanup quiescence), §11.4.161 (rootless / scoped `CAP_NET_ADMIN` exception), §1.1 (paired mutation) — from `CLAUDE.md` in-context.
- Phase-0 spike rig **[04_P0 §3]** + WBS items HVPN-P0-022/023 cited via the overview — **not independently re-fetched** (`UNVERIFIED` for exact item numbering; the rig design is reproduced from the overview skeleton).